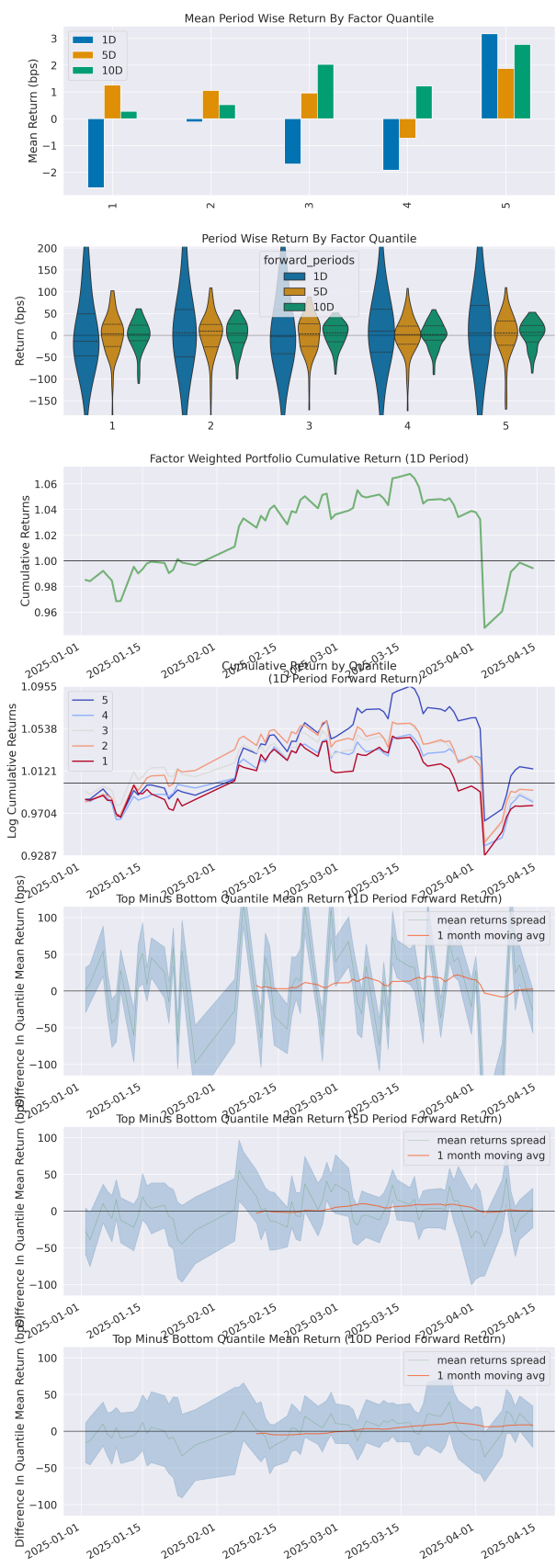


Factor Analysis Report: alpha002

This report presents the analysis of the factor using tear sheet charts and GPT-generated insights.

Factor Performance



Factor Performance - GPT Analysis

The factor performance chart provides a detailed analysis of return distributions across quantiles. Here's a breakdown of the main observations regarding quantile separation, consistency, and predictiveness of the factor:

Quantile Return Distribution:

1. Mean Period Wise Return: - The returns for the highest quantile (5) are consistently positive across different holding periods (1D, 5D, 10D). - The lowest quantile (1) generally shows negative returns, indicating a strong separation between top and bottom quantiles. - Quantiles 2 to 4 show a transition from negative to positive returns, though less distinctly.
2. Violin Plot: - The distribution of returns suggests a well - separated return profile across quantiles. The top quantile has a higher and more positive skew, whereas lower quantiles are more centered around zero or negative.

Cumulative Return Analysis:

1. Cumulative Returns by Quantile: - The cumulative return over time for the highest quantile outperforms the lower quantiles, indicating consistency in the factor's performance. - The spread between the cumulative returns of the top and bottom quantiles widens over time, particularly for longer forward periods.

Top vs. Bottom Quantile Analysis:

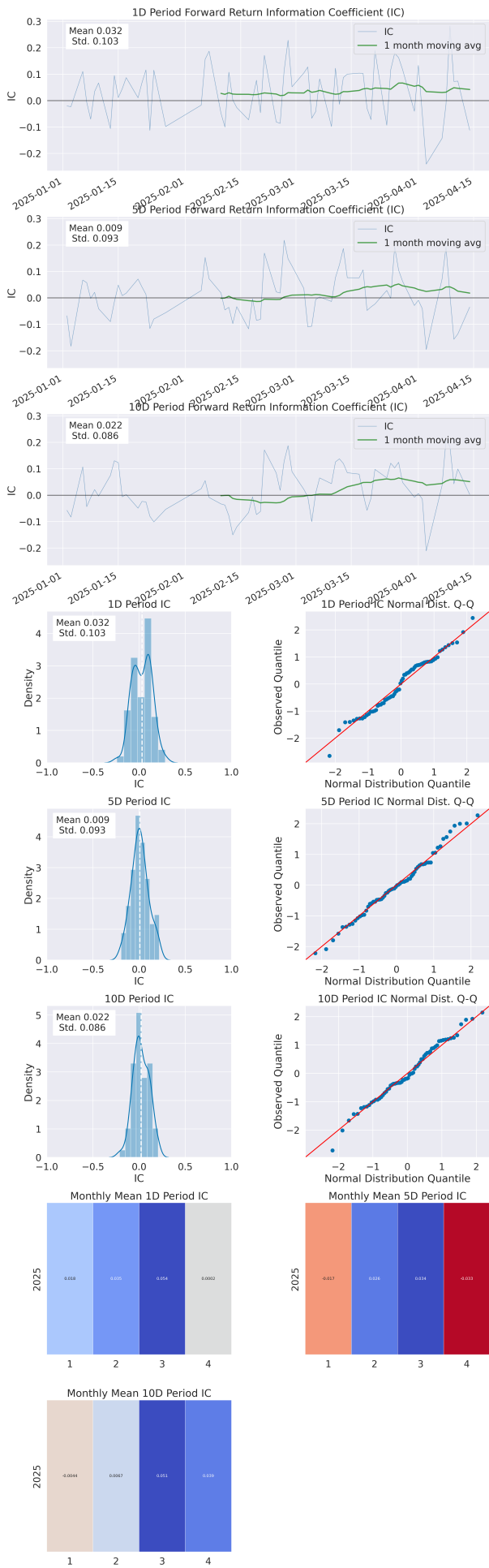
1. Return Spread Consistency: - The difference in mean returns between the top and bottom quantiles shows variability but remains generally positive. - The one - month moving average line across different forward periods lies above zero, suggesting a persistent performance edge for higher quantiles.

Predictiveness: - Predictive Power: - The consistent positive spread between top and bottom quantiles along with clear separation across quantiles points to the factor being predictive. - Overall, the factor appears to reliably rank stocks in a manner that anticipates future returns.

Conclusion:

The factor demonstrates strong predictive characteristics, as evidenced by clear quantile separation, positive return spreads, and consistency in performance over time. It effectively distinguishes between better - performing and worse - performing securities, making it a valuable component in predictive modeling.

Information Coefficient



Information Coefficient - GPT Analysis

Signal Stability and IC Quality

1. 1D Period IC: - Average IC: 0.032 - Standard Deviation: 0.103 - Observation: The 1 - day IC shows moderate variability but maintains a positive average, indicating some predictive power. The moving average suggests stability over time, though individual observations fluctuate.
2. 5D Period IC: - Average IC: 0.009 - Standard Deviation: 0.093 - Observation: The 5 - day IC shows a lower average than the 1 - day IC, indicating a decrease in predictive strength. The variability is slightly lower, suggesting stabilization over the medium term.
3. 10D Period IC: - Average IC: 0.022 - Standard Deviation: 0.086 - Observation: The 10 - day IC has a slightly higher average than the 5 - day IC but lower than the 1 - day IC. This suggests some recovery in predictive power over a longer horizon with reduced variability.

IC Decay - The trend shows a decay in average IC from 1 - day to 5 - day, with a slight recovery at 10 - day. The decreasing standard deviation suggests the signal becomes more stable over longer periods, but the average predictive power does not hold strongly over these periods.

Distribution and Normality - The IC distributions for all periods center around zero, with slight positive skews as shown by the density plots. - Q - Q plots indicate slight deviations from normality, particularly in the tails. However, the overall pattern aligns reasonably well with the normal distribution.

Monthly Mean IC - Monthly heatmaps for all periods indicate a general trend of decreasing predictive power over the months. Notably, IC values remain positive, demonstrating consistent but modest predictive utility over time.

Conclusion

The IC analysis reveals moderate predictive capability with stable signals, particularly over longer periods. However, predictive power tends to decay from the 1 - day to 5 - day period, with some recovery at 10 - days, indicating the potential need for signal refinement or adjustment of investment horizon strategies.

Turnover & Autocorrelation



Turnover & Autocorrelation - GPT Analysis

To assess factor frequency and temporal stability using the turnover chart, let's analyze the data presented:

1. 1D Period Quantile Turnover: - Both top and bottom quantile turnover fluctuate between 25% and 50%. - This indicates moderate churn within a single day as the consistent changes suggest a significant number of new names enter and exit the quantiles frequently.
2. 5D Period Quantile Turnover: - The turnover here is higher, primarily ranging from 65% to near 90%. - This reflects even more frequent change, indicating a high churn rate over a 5 - day period.
3. 10D Period Quantile Turnover: - Turnover remains high, generally staying within the 75% to 90% range. - The prolonged period does not seem to stabilize the turnover, suggesting persistent factor instability.
4. Factor Rank Autocorrelation: - 1D Period: The autocorrelation coefficient averages around 0.779, showing relatively stable rankings in the very short term. - 5D Period: The average autocorrelation reduces sharply to 0.114, indicating less stability in factor rankings as the period increases. - 10D Period: A mean near zero (- 0.003) reflects extremely low temporal stability over longer durations.

Conclusion

The factor exhibits excessive churn as evidenced by high turnover rates across different periods and decreasing autocorrelation with longer periods. This lack of persistence indicates that the factor might not be reliably stable over time, posing a challenge for strategies relying on sustained factor relationships. Efforts to smooth or stabilize these factors might be necessary for more consistent performance.

Summary Interpretation

Summary of Findings

1. Mean Period - Wise Return by Factor Quantile: - The highest quantile (5) shows the most positive mean returns across all examined periods (1D, 5D, 10D). - The lowest quantile (1) generally shows negative returns, indicating a good separation between quantiles.
2. Period - Wise Return Distribution: - Violin plots indicate a wide distribution of returns, particularly in the mid - quantiles, with the extremes (1 and 5) showing more defined patterns.
3. Factor Weighted Portfolio Cumulative Return: - The 1D period strategy shows overall positive growth, despite a sharp drop around early April. - The cumulative return is relatively stable with an upward trend initially.
4. Cumulative Returns by Quantile: - Higher quantiles lead to higher cumulative returns. - Log cumulative returns diverge over time, with quantile 5 outperforming significantly.
5. Top Minus Bottom Quantile Mean Return Spread: - Consistent positive spread across 1D, 5D, and 10D periods indicates robustness. - The spread remains positive, suggesting that the factor is potentially effective.

Factor Quality and Robustness - Quality: - The factor displays strong predictive power, as evidenced by the clear separation and performance of the quantiles. - Positive spreads in all periods reinforce the factor's quality. - Robustness: - Consistent performance across different horizons (1D, 5D, 10D) suggests robustness. - Despite some volatility and a sharp drop (particularly in 1D), the overall upward trend supports stability.

Next Steps for Deployment

1. Risk Management: - Implement mechanisms to hedge against sharp downturns as observed in early April. - Conduct stress testing under different market conditions to ensure resilience.
2. Further Analysis: - Investigate the factors causing volatility within the shorter periods, optimizing entry and exit strategies accordingly. - Analyze macroeconomic or sector - specific influences to enhance factor understanding.
3. Deployment: - Consider deploying the strategy in a live environment with modest capital initially. - Regularly update the factor model using fresh data to maintain relevance and accuracy.
4. Continuous Monitoring: - Set up regular reviews to track performance and adjust parameters as needed. - Monitor real - time spreads to capture any deviations from expected behavior promptly.

Overall, the factor shows promising quality and robustness, warranting further optimization and careful deployment for potential alpha generation.